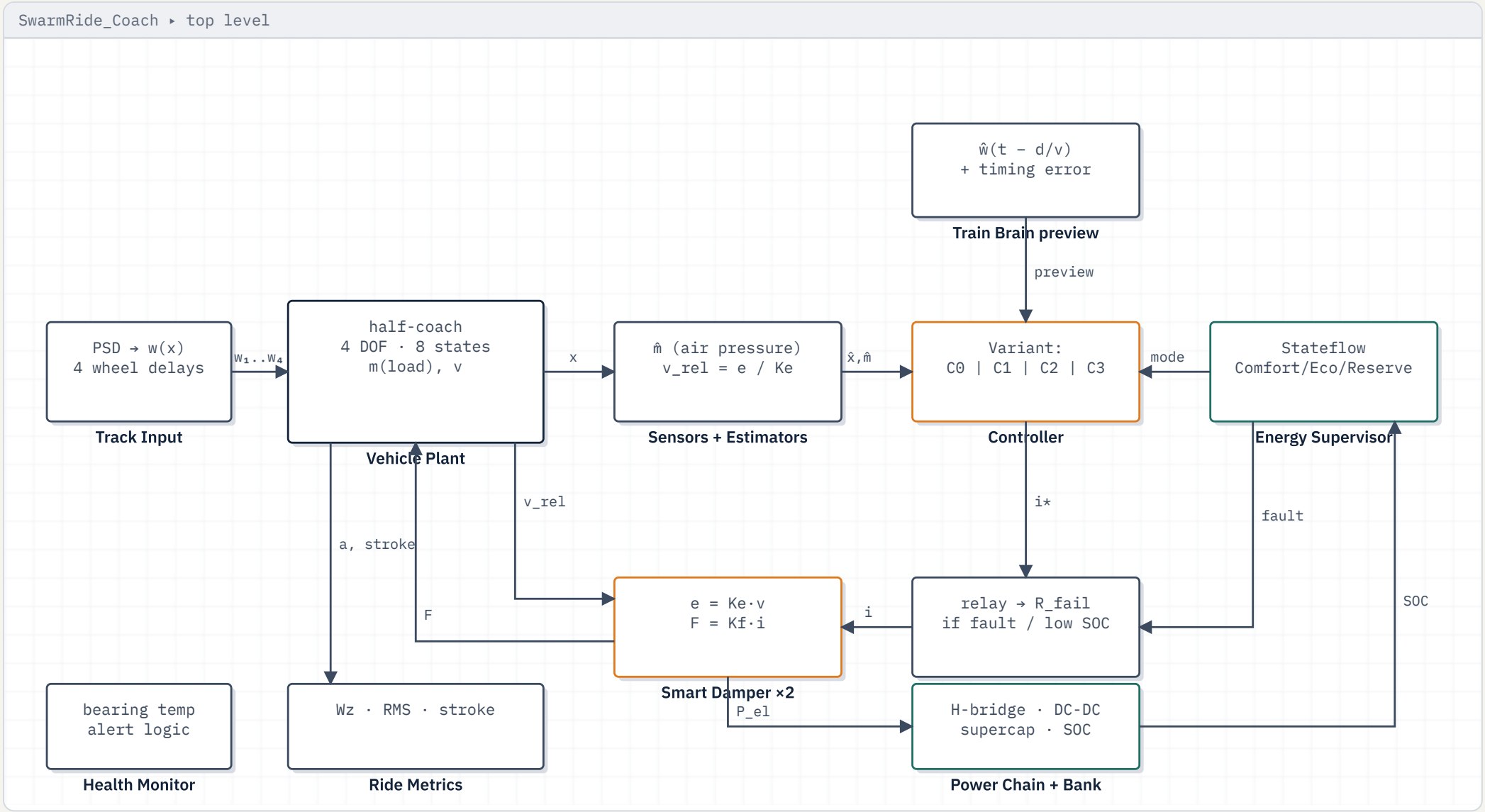
What the MATLAB/Simulink model looks like

One model, one track, four controllers. Every result on the next pages comes from running this structure; nothing is drawn by hand.



Same track for every case, so differences come only from the damper.

Variant controller: C0 oil damper · C1 fixed-load · C2 SwarmRide · C3 + preview.

Everything logged: ride, stroke, force, power and charge feed the plots.

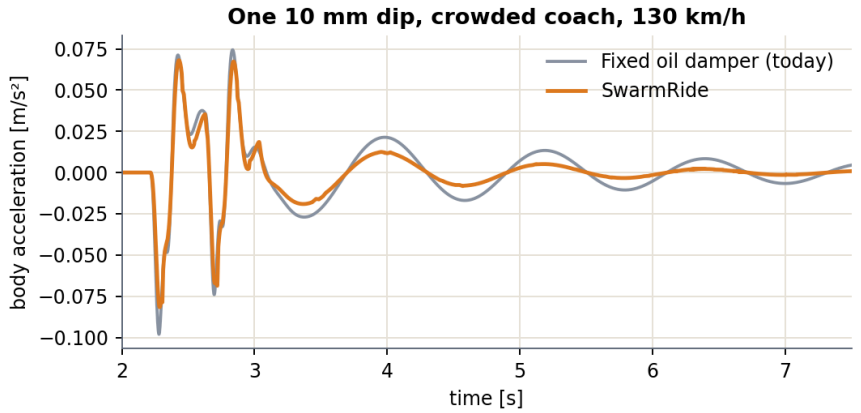
Preliminary results on pages 2–4 come from a Python version of this exact model with assumed coach data. The MATLAB/Simulink build replaces them.

ESSENTIAL RESULTS 1–4

A smoother ride than today's dampers

1 • One bump: the bounce dies faster

PASS

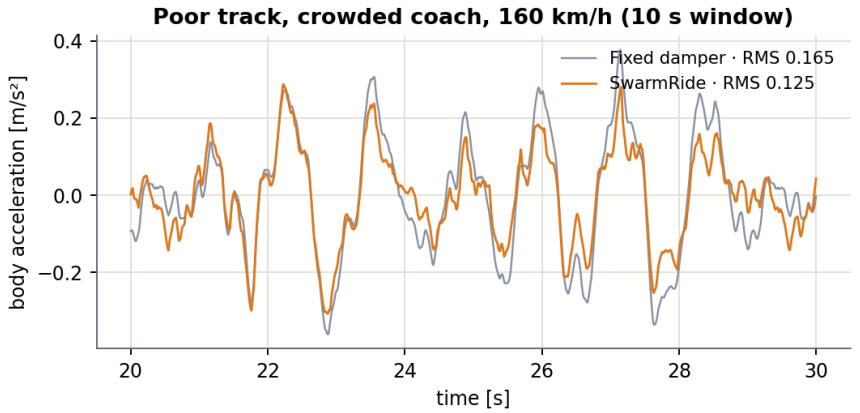


Proves: after a 10 mm dip, peak body acceleration drops from 0.10 to 0.08 m/s² (17% lower) and the bounce settles sooner.

Pass condition: lower peak and faster settling than the oil damper.

2 • Real track: less shaking, all the time

PASS

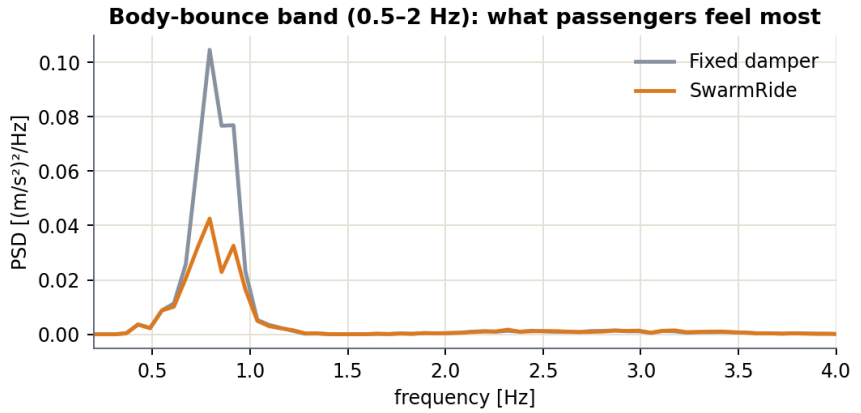


Proves: on poor track at 160 km/h with a crowded coach, body vibration falls from 0.165 to 0.125 m/s² RMS (24% lower).

Pass condition: lower RMS acceleration on identical track.

3 • It removes the vibration people feel

PASS

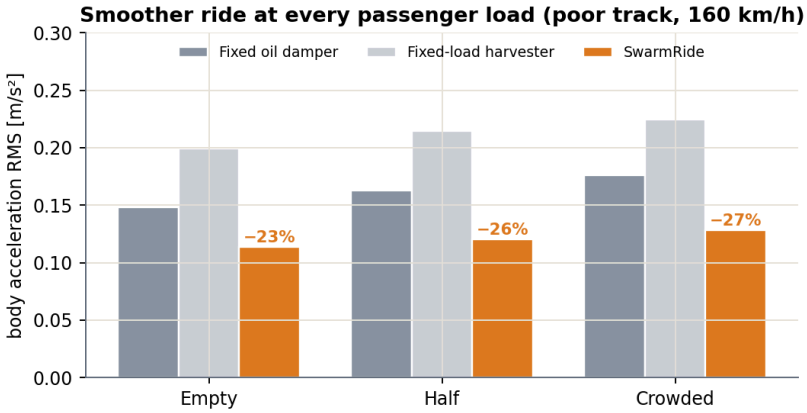


Proves: in the 0.5–2 Hz band passengers feel most, vibration falls 30% (0.157 → 0.110 m/s²). Above 6 Hz it rises slightly (0.007 → 0.009 m/s²) from fast damping changes; Simulink adds command smoothing.

Pass condition: lower vibration in the 0.5–2 Hz band.

4 • Better at every passenger load

PASS



Proves: vibration is 23%, 26% and 27% lower for empty, half and crowded coaches. Ride index Wz: 1.90 → 1.69 when crowded.

Pass condition: better than both the oil damper and a fixed-load harvester at every load.

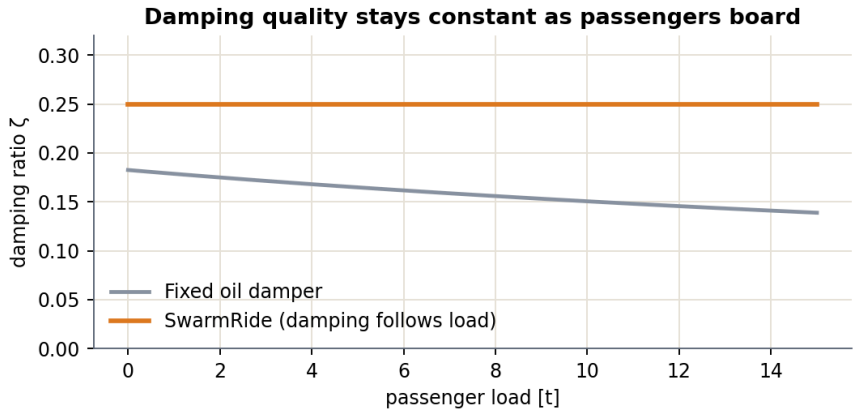
Poor track = high-irregularity track spectrum. Coach parameters assumed (Guide, Section 8).

ESSENTIAL RESULTS 5–8

Regeneration into the power bank

5 • Damping follows the passenger load

PASS

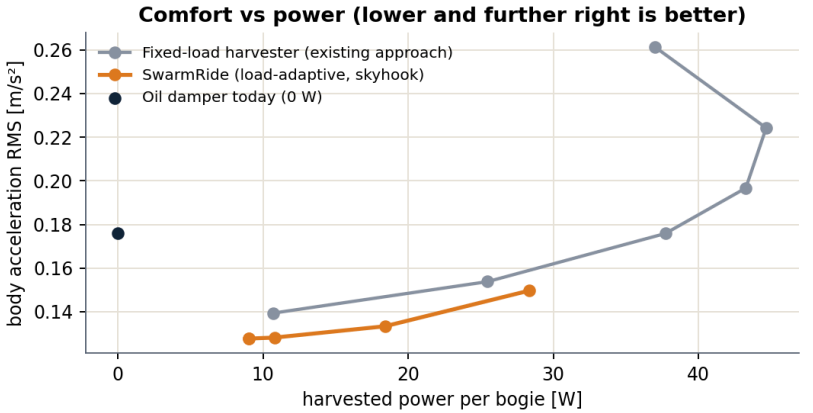


Proves: a fixed damper loses damping quality as people board; SwarmRide holds it at the chosen target.

Pass condition: constant damping ratio from empty to crowded.

6 • Comfort and power together

PASS

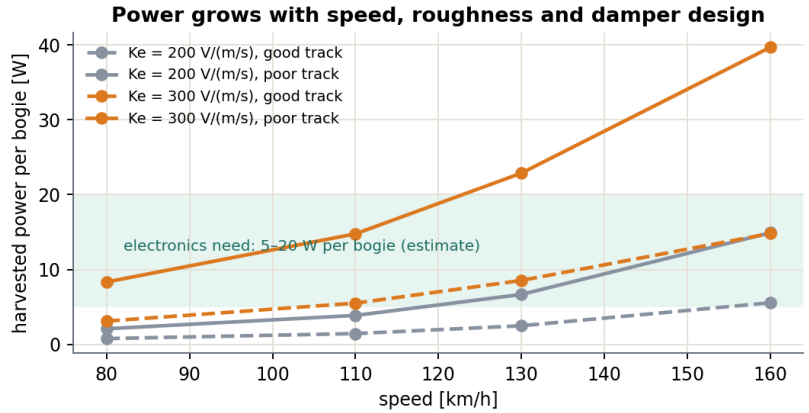


Proves: at the same 10–28 W, SwarmRide always rides smoother than a fixed-load harvester. Fixed-load harvesters reach 45 W only by making the ride 28% worse than today's oil damper.

Pass condition: SwarmRide curve below the fixed-load curve at equal power.

7 • How much power, and how to get more

NEEDS WORK

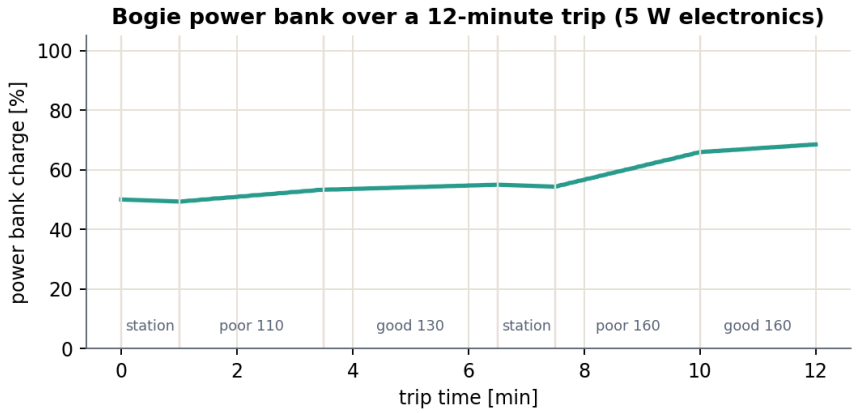


Proves: at 160 km/h: 6 W (good track) to 15 W (poor track) per bogie with today's damper design; a stronger magnet/coil (Ke = 300) gives 15 to 40 W.

Pass condition: harvest above the electronics need on typical routes.

8 • The bogie keeps itself charged

PASS



Proves: over a 12-minute trip with station stops, the power bank ends at 68% with 5 W electronics and the Ke = 300 design.

Pass condition: charge never hits zero; ends at or above the start.

All power figures per bogie, after converter losses (92% assumed). Electronics need is an estimate until the bench prototype is measured.

SCORECARD

Where the approach stands today

Claim	Preliminary result	Status
Smoother ride on a bump	17% lower peak	PASS
Smoother ride on real track	24% lower RMS (poor, crowded, 160 km/h)	PASS
Better at every load	23–27% lower RMS, empty to crowded	PASS
Better than existing harvesters	SwarmRide curve beats the fixed-load curve	PASS
Purely regenerative (never pushes)	$F \cdot v_{rel} \geq 0$ in every step (minimum 0.0)	PASS
Small high-frequency side effect	above 6 Hz: 0.007 → 0.009 m/s² (smooth the damping command)	NEEDS WORK
Stroke within limits	max 12.0 mm vs 13.4 mm today (limit ±40 mm assumed)	PASS
Power above electronics need	good track 1–6 W (Ke 200) · 3–15 W (Ke 300)	NEEDS WORK
Power bank stays charged	ends trip at 68% (Ke 300, 5 W electronics)	PASS
Bump preview (C3) adds comfort	to be built in Simulink (week 6)	NEXT
Fail-safe equals passive damper	to be built in Simulink (week 5)	NEXT

TO FIX THE POWER GAP

1. Stronger damper: more magnet and coil turns (higher Ke), lower coil resistance.
2. Low-power electronics: sensors sampled in bursts, radio duty-cycled (5 W target).
3. Harvest lateral and yaw dampers too on Vande Bharat bogies.
4. Supercap + small battery sized for the longest station stop.

SHOWN TO INVESTORS AS

1. Side-by-side animation: oil damper vs SwarmRide on the same track.
2. Three headline numbers: ride improvement, watts per bogie, power-bank charge.
3. This scorecard, updated after Simulink and the bench test.
4. Live dashboard built from the exported simulation results.

Preliminary: simplified model, assumed parameters, bump preview not yet included. Final numbers come from the MATLAB/Simulink study and the bench prototype.